UVAMT 2026

Opening Ceremony

Will commence at 10:00am

UVAMT 2026

Opening Ceremony

Welcome to all!

Schedule

10:30am: Individual Round (45 min)

11:30am: Team Round (30 min)

12:00pm: Lunch + Professor Panel

1:15pm: Optimization Round (40 min)

2:10pm: Game Round

3:00pm: Tiebreakers, Awards, Closing Ceremony

4:30pm: Event Ends

Contestants

- 100+ contestants, ranging from Northern Virginia to Charlottesville
- 22 teams total

Prizes

Medals and trophies for:

- Top 5 individuals
- Top 3 teams
  - Ranking: (3 highest Individual Round scores) + 2*(Team Round score)
- Assorted smaller prizes (e.g. water bottles) for Optimization and Game rounds

Professor Panel

- Held during lunch (12-1pm) in this room (402)
- Competitors and chaperones welcome!
- Ask about:
  - UVA's Math, Statistics, CS programs
  - Teaching, research, etc
  - Or just listen!

Professor Panel

- Panelists:
  - Prof. Shounak Chattopadhyay (Statistics)
  - Prof. Raymond Pettit (Computer Science)
  - Dr. Sarah Blackwell (Mathematics)
  - Eleftherios Chatzitheodoridis (Mathematics, PhD)
  - David Chasteen-Boyd (Mathematics, Graduate)

Answer Format

- Individual and Team: **numerical answers**
  - Integer answers between -1,000 and 1,000 must be expressed as such
  - Fractional answers with numerator and denominator between -1,000 and 1,000 must be expressed either as a fraction in lowest terms or an **exact** decimal
  - Denominators do not have to be rationalized
  - Unreasonable calculations (e.g. 2^20, sin(π/16)) can be left as-is, but simple calculations must be carried out (e.g. 2+1/3, sin(π/4))
- Check your work!

Note On Difficulty

- Individual and Team: problems are in **difficulty order**
- Meant to be challenging
  - it is **normal** to only solve a few problems!

Questions?

- Let us know if you have any questions - otherwise, good luck!

Individual Round

- 15 problems, 45 minutes
- Numeric answers
- Write full name on answer sheet
- To be worked on individually
    - Prohibited: calculators, phones, any communication

Team Round

- 15 problems, 30 minutes
- Numeric answers
- Write Team ID and all names on answer sheet
- To be worked on collaboratively within teams
    - Prohibited: calculators, phones, communication with anyone outside your team's contestants

Team Round Room Assignments

- Teams 01, 02, 03, 04:
  - Room 244
- All other teams:
  - Room 325

Lunch + Panel

- Pizza outside on the 3rd floor
- Professor panel in room **402**
- Room **325** will remain open for lunch, as well as outdoor space
- Be back in your assigned room (325 or 244) by **1:00pm**

Optimization Round

- 3 problems, 40 minutes
  - Mid-round submission at 25 minutes remaining
- Problems do not have a single correct answer; your goal is instead to minimize/maximize the quantity indicated
- To be worked on collaboratively
  - Prohibited: calculators, phones, communication with anyone outside your team's contestants

Optimization Round

- Mid-round submission:
  - Submit at 25 minutes left
  - Receive graded submission at 10-15 minutes left
  - Verdicts: ✓ or "INVALID", followed by score or reason
- End-of-round submission:
  - If not improved from mid-round, we'll use your mid-round submission

Optimization Round

- Determining a winner:
  - For each problem, you'll be ranked first on your 40-minute submission, then on your 15-minute submission
  - The winning team is the team with the lowest *product* of ranks
- Winners for each individual problem also recognized

Optimization Room Assignments

- Teams 01, 02, 03, 04:
  - Room 244
- All other teams:
  - Room 325

Team Assignments

Team ID	Chaperones	Team Members					
01	Tracy Lou	Jordon Sanders	Katelyn Kerns	Shola Amuda	Natalie Miller		
02	Liz Larson	Annie Robertson	Nati Yohannes	Zeke Lewis	Aoife Keup		
03	Christine Le	Richard Phan	Harry Han	Kevin Chen	Isaac Park		
04	Emelie Skuce and Violet Raftery	Avery Uhle	Sage Anastasia	Autumn Robinson	Matthew Muth	Ninghao Zhan	
05	Tracy Lou	Makayla Hagstrom	Anna Varkonyi	Arthur Molnar	Steph Cuprak	Marek Cook	
06	Eucee Chan	Ryan Chan	Rudy Khanwalkar	Wilson Cao	Aryan Raj	Victor Moldoveanu	
07	Tracy Lou	Jonas Westerman	Natalie Bills	Dilorom Shukhratova	Adrian Ramos	Ayden Eusanio	
08	Patrick Kennedy	Abbos Yusopov	Alex Tan	Vihaan Shirodkar	Aaradhya Shetty	Olivia Zhang	
09	Tracy Lou	Karina Kaczmarczyk	Tyler Conway	Aline Mambourg	Madelyn Zealor	Ben Jones	
10	Tracy Lou	Hameda Rezaie	Sabera Rezaie	Elizabeth Bonilla	Cooper Cuyler	Adelaide Cook	
11	Anthony Armbrister	Dersabel Gebrengus	Kavya Vajjhala	Aissatou Sall	Ange Desiree Monkam Younga	Christabel Ellis	
12	Melissa Saunders, Donald Burd, David Woodward	Evelyn Saunders	Emma Somers	Josiah Greek	Junhao Zheng	Serene Song	Serah Song
13	Liz Larson, Sonya Stevens, Razieh Shamsedin	Addie Thompson	Peter Dumars	Eli Hoover	Jayse Mcfadden	Parsa Motaharinezhad	Dorsa Motaharinezhad
14	Benjamin Peck	Noah Peck	Felix Mason	Owen Paulson	Eshan Panicker	Zaki Farhat	Phoebe Fischman
15	David Woodward	Nate Paul	Audrey Chen	Andrew Wang	Eshaan Mital	Alex Chen	Philippe Deur
16	Arun Thota	Anish Thota	Aarush Gupta	Srinandasai Ari	Kossara Tabakova	Ashvik Mahapatra	Vincent Chu

- Teams 01, 02, 03, 04: Room 214
- All other teams: Room 325

Game Round

- 3 main rounds involving dice
  - Each team's submitted dice will be pitted against each other
  - To be worked on collaboratively within teams
  - Cooperation between teams is encouraged!
- 2 break rounds in between
- Low stakes, meant to be fun

Game Round #1

Each team must submit one 6-sided die.

Constraints

- Each face must be an integer from 1-6
- The sum of the faces must equal 21.

Tournament Rules:

- Every pair of dice plays a 100-round match
- Each round: both dice are rolled
- Higher number wins (Win: +1, Loss: -1, Tie: 0)

Each team plays every other team exactly once. Final score is the sum of all match points.

Game Break #1

Guess the Mode

Each team submits one number from 1–100.
The volunteers also choose one number.

After all submissions, we compute the mode
(the number chosen by the most teams).

Rules

- If the volunteers' number matches the mode, no prize.
- If the mode is not unique, no prize.
- Otherwise, teams that chose the mode win candy

Name	Score		Name	Score		Name	Score		Name	Score
4A	130		4B	47		14B	-35		5A	-106
1B	128		11A	40		3A	-36		15A	-106
13B	126		7A	26		8A	-41		10A	-107
2A	106		3B	7		15B (Alben	-45			
12A	94		10B	-5		7B	-59			
13A	77		5B	-12		6A	-60			
11B	72		9C	-16		16A	-71			
9A	70		12B	-20		2B	-72			
6B	58		1A	-25		9B	-80			
16B	48		14A	-33		8B	-100			

Game Round #2

Round 1 scores have now been released.

Based on these results, each team may **submit a new die.**

But there is a twist...

Claude will also play in this round.

Instead of submitting one die, Claude submits many dice, sampled **uniformly at random from all valid dice**.

Your new die will compete against:

- **All other teams' dice**
- **The randomly generated dice from Claude**

Game Break #2

Guess the Standard Deviation

Each team submits one number from 0–100 (decimals are allowed).

After all submissions, we compute the standard deviation (formula provided on the right).

Try to be as close to the standard deviation as possible!

Prizes

- Top 3 closest teams win candy!

Population Standard Deviation

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$$

Where:

- x_i = each data value
- μ = population mean
- N = number of data points
- σ = population standard deviation

Name	Score		Name	Score		Name	Score		Name	Score
6B	12672		5A	10724		7B	8755		3B	3691
4A	12535		16A	10567		15A	8229		8A	3687
10B	12383		1A	10171		10A	8154			
4B	12266		14A	10149		11A	7739			
9B	11422		16B	10107		12B	7202			
9A	11198		5B	10102		11B	7182			
1B	11097		14B	10032		13A	5878			
13B	11057		8B	9842		6A	3963			
2A	11055		9C	9016		15B (Alben	3915			
12A	10962		7A	8962		2B	3911			

Game Round #3

Round 2 scores have now been released.

Based on these results, each team may **submit two new die.**

Updated Rules:

- Faces must be positive integers from 1-6
- Total sum of faces across both dice is **30**
- Greater *product* wins
- **Claude is still competing!**

Example

- Team 1 rolls (3, 3) and Team 2 rolls (2, 5)
- Team 2 wins as 2 * 5 > 3 * 3

14A with Score 15154

3 3 3 3 3 3 | 2 2 2 2 2 2

6B with Score 14995

3 3 3 3 3 3 | 2 2 2 2 2 2

14B with Score 14848

2 2 2 2 2 2 | 3 3 3 3 3 3

Name	Score		Name	Score		Name	Score
14A	15154		9B	9771		11A	4996
6B	14995		16B	9281		11B	4974
14B	14848		15A	8676		8A	3236
15B (Albem	13684		1B	8328		7A	998
4B	13675		5B	7119		13A	-2255
5A	13152		16A	6902		10A	-4005
6A	12311		8B	6274		12B	-4794
10B	11750		13B	6238		9C	-5846
1A	10613		2A	6090		12A	-7561
4A	10057		7B	5184			

UVAMT 2026

Closing Ceremony

Starts 3:25pm

Our Helpers

- **Core Team**: Mikhail Kornilov, Liam Baird, Vincent Trang, Utkarsh Goyal, Emma Ylagan, Isaac Assink, Brendan Malaugh, Eric Zeng
- **Student Helpers**: Vrishak Vemuri, Thomas Kennedy, Cathy Smolka, Jacob Ezieme, Julian Cerineloo, Mayna Malhotra
- **Panelists**: Prof. Shounak Chattopadhyay, Prof. Raymond Pettit, Prof. Sarah Blackwell, David Chasteen-Boyd, Eleftherios Chatzitheodoridis
- **Faculty Helpers**: Professor Foldes, Professor Mark, Dallas Pitt, Melissa Kollar

Sponsors

Gold Tier: Jane Street

Silver Tier: HRT®

Bronze Tier: SPARC Athemath

Sponsors - Jane Street

- What is Jane Street?
  - Quantitative trading firm
    - Trading on financial markets
  - Emphasizes:
    - Collaboration
    - Puzzle / game theory ethos

Jane Street

Sponsors - Jane Street

- Summer internship:
  - 11 weeks, May - August
  - For college juniors + sophomores
  - Roles: trading, research, SWE
  - No prior financial knowledge required
  - Very challenging but fun experience!

Jane Street

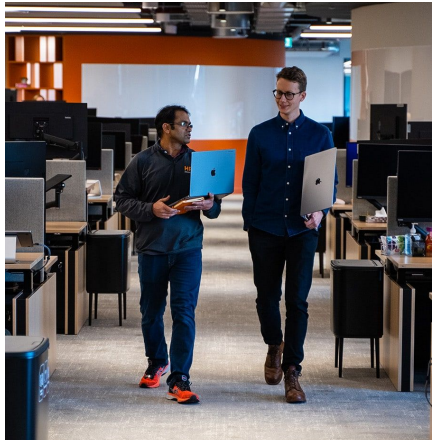
Sponsors - Jane Street

- Other programs:
  - FTTP (for college freshmen)
  - FOCUS, AMP, INSIGHT (for underrepresented groups)

Jane Street

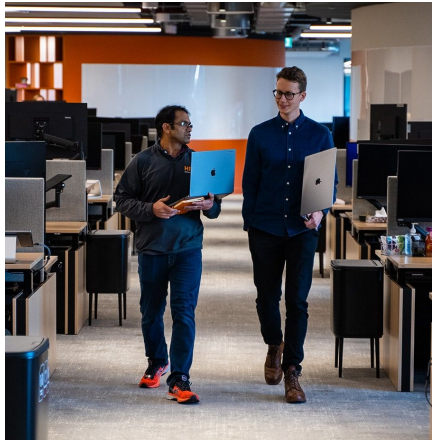
Sponsors - Hudson River Trading

- What is HRT?
  - Quantitative trading firm
    - Trading on financial markets
  - Emphasizes:
    - Collaboration
    - Algorithmic / engineering ethos



Sponsors - Hudson River Trading

- Programs:
  - Summer internship (11 weeks)
    - Roles: software engineering, algorithm development, algorithm trading
  - Winter internship (2-4 weeks)
    - Compressed version of summer internship

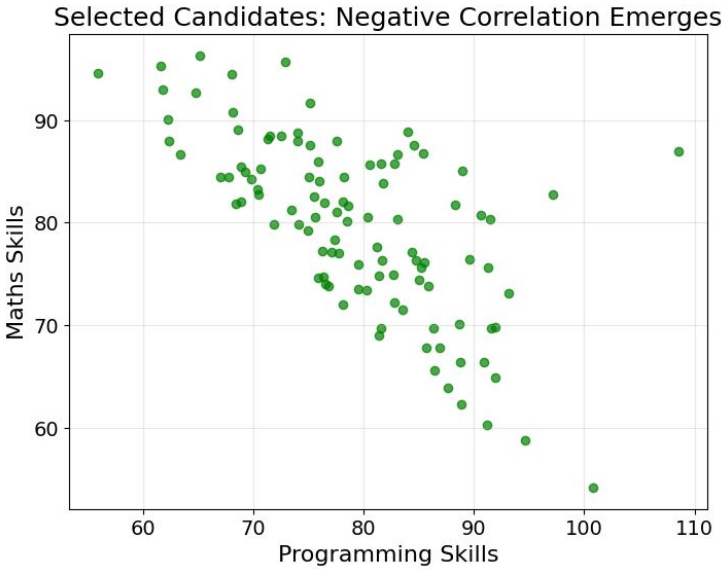


Sponsors - SPARC

SPARC

Sponsors - SPARC

Berkson's Paradox



SPARC

Sponsors - SPARC

Improv Monopoly

SPARC

Sponsors - SPARC

You can just do things

SPARC

Sponsors - Athemath

- What is Athemath?
  - Online math program for middle/high school girls
    - Non-binary also welcome!
  - High-quality classes, office hours, community activities, meetups
  - Apply at: athemath.org/apply

Athemath

Sponsors - Athemath

Not just a math class:

A camp-like experience,
for 12 weeks.

Saturday	Saturday (Evening)	Sunday (Morning)	Monday (Evening)	Tuesday (Evening)
Class w/ Teacher	*Live Office Hours*	*Performing Arts Club*	*Chess Club*	*Seminar on Olympiads*
Tuesday (Evening)	**Wednesday (Evening)**	**Thursday (Evening)**	**Friday (Evening)**	**Friday (Evening)**
Painting w/ TA's	*Japanese Learning Club*	*Games w/ TA's*	*Seminar on Primes*	*Coffee with TA's*

Fall and **Spring**

Athemath

Sponsors - Athemath

Top mathletes as staff:

Board

Greta
MOP
USAJMO x2

Emily
USAMO
MOP x2

Vivian
EGMO
MOP x3

Patrick
IMO
MOP

Catherine
MOP

Vera
CGMO

Rachel
USAJMO
CMO

Andrew
MOP

Teachers

Amy Cui **MOP** x2 Ana Boiangiu **EGMO** **MOP** Ashley Zhu **MOP** Catherine Xu **MOP** Ekam Kaur **MOP** Emily Liu **MOP** Greta Qu **MOP** Jiya Dani **MOP** Julia Xiang **MOP** Matilde Iannaccone **MOP** Pimmy Maneepairoj **MOP** Sophia Jin **MOP** Vivian Loh **EGMO** **MOP** x3 Wendy Leong **CGMO** **MOP**

TAs

Akshitaa Arun **INMO** Andrew Xie **MOP** Angeline Zhao **USAMO** Jingni Liao **Aus MO** Laura Nan **IMO** **EGMO** Lillian Xu **USAMO** Neal Yan **MOP** Peyton Li **MOP** Rachel Chen **USAJMO** Ritwin Narra **MOP** Swastika Deb Uddipa Pal Tayakorn Suwanich **IMO** **MOP** Vera Puah **CGMO** Yash Vardhan **IMOTC**

Athemath

Sponsors - Athemath

Giant Athemath meetup!!

Athemath

Prizes

- What we're here for!

Optimization Round

Optimization (Q1)

- Question 1 winner:

Optimization (Q1)

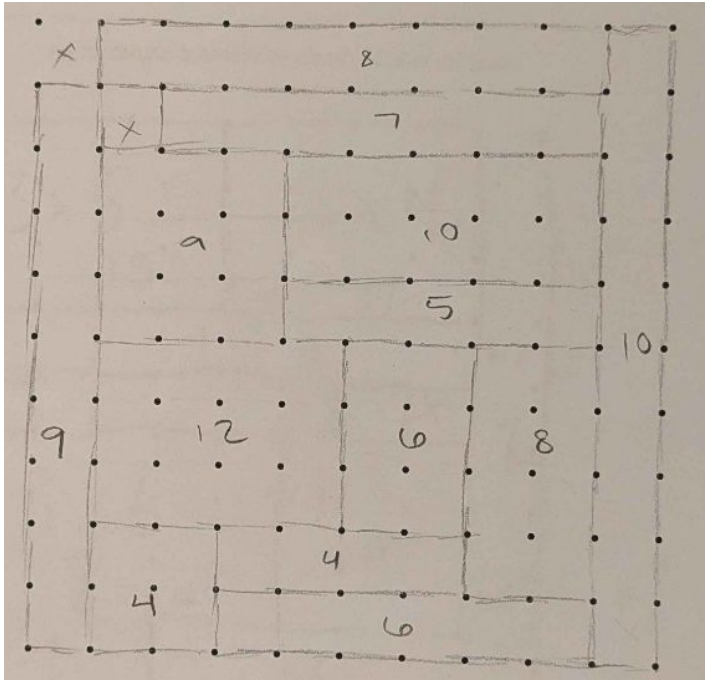
- Question 1 winner:
  - Team 06
  - Score: 2.401 x 10^865
  - Expression: (1+6)^(8^3 x 2) = 7^(4^5)

Optimization (Q2)

- Question 2 winner:

Optimization (Q2)

- Question 2 winner:
  - Team 08
  - Score: 8



Optimization (Q3)

- Question 3 winner:

Optimization (Q3)

- Question 3 winner:
  - Team 06
  - Score: 2/13 = 0.1538

	2	1	1	1	2	
	1		1		1	
	1	1	2	1	1	
	1		1		1	
	2	1	1	1	2	

Optimization (Overall)

- 3rd place:

Optimization (Overall)

- 3rd place:
  - Ranks: 2nd x 4th x 5th = **40**

Optimization (Overall)

- 3rd place:
  - Ranks: 2nd x 4th x 5th = **40**
  - **Team 15**

Optimization (Overall)

- 2nd place:

Optimization (Overall)

- 2nd place:
  - Ranks: 2nd x 2nd x 5th = **20**

Optimization (Overall)

- 2nd place:
  - Ranks: 2nd x 2nd x 5th = **20**
  - **Team 03**

Optimization (Overall)

- 1st place:

Optimization (Overall)

- 1st place:
  - Ranks: 1st x 6th x 1st = **6**

Optimization (Overall)

- 1st place:
  - Ranks: 1st x 6th x 1st = **6**
  - **Team 06**

Optimization (Overall)

- Our testsolvers have spent many hours perfecting their own solutions to these three problems!
  - Feel free to talk to them!

Overall Top Teams

- Ranking: (3 highest Individual Round scores) + 2 * (Team Round score)

3rd Place Team

3rd Place Team

- Team 16
  - Members: Anish Thota, Aarush Gupta, Srinandasai Ari, Kossara Tabakova, Ashvik Mahapatra, Vincent Chu
  - Team Score: 7
  - Top 3 Individuals Total Score: 20

2nd Place Team

2nd Place Team

- Team 06
  - Members: Ryan Chan, Rudy Khanwalkar, Wilson Cao, Aryan Raj, Victor Moldoveanu
  - Team Score: 8
  - Top 3 Individuals Total Score: 24

1st Place Team

-

1st Place Team

- Team 15
  - Members: Nate Paul, Audrey Chen, Andrew Wang, Eshaan Mital, Alex Chen, Philippe Deur
  - Team Score: 10
  - Top 3 Individuals Total Score: 24

Individual Round

- There were multiple sets of ties - we will call people up for tiebreakers!

Tiebreakers

- All tied contestants are given the same problem at the same time
- Raise your board to submit: first correct answer wins!
- Remaining contestants compete for remaining ranks (until 1st-5th are all determined)
- 3 minute time limit

Ranks 4-8

Ranks 4-8

- Score: 7

Ranks 4-8

- Score: 7
- Competitors: Abbos Yusopov,

Ranks 4-8

- Score: 7
- Competitors: Abbos Yusopov, Anish Thota,

Ranks 4-8

- Score: 7
- Competitors: Abbos Yusopov, Anish Thota, Felix Mason,

Ranks 4-8

- Score: 7
- Competitors: Abbos Yusopov, Anish Thota, Felix Mason, Kossara Tabakova,

Ranks 4-8

- Score: 7
- Competitors: Abbos Yusopov, Anish Thota, Felix Mason, Kossara Tabakova, Ryan Chan

Tiebreaker 1.1

Tiebreaker 1.1

A regular tetrahedron with volume 1 is cut into two pieces by a plane parallel to one of its sides passing through its center. What is the volume of the smaller piece?

Tiebreaker 1.2

Tiebreaker 1.2

Vincent is playing a counting game. He starts at 1, then counts up by adding the same positive integer each time. How many different ways can he do this so that 25 appears in his sequence at some point?

Tiebreaker 1.3

Tiebreaker 1.3

Evaluate $\displaystyle\sum_{n=-4}^{4} \frac{1}{3^n + 1}$.

Ranks 2-3

Ranks 2-3

- Score: 9

Ranks 2-3

- Score: 9
- Contestants: Alex Chen, Nate Paul

Tiebreaker 3.1

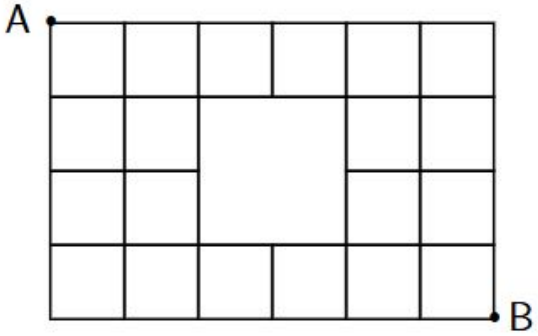
Tiebreaker 3.1

Four pushpins are pinned into a corkboard to mark the corners of a unit square. A *move* denotes removing exactly two pushpins and reinserting them so that the four pushpins again form the corners of a square (of any size or orientation). Suppose that one of the pushpins started at point P, and after three moves, ended at point Q. Find the maximum possible distance between P and Q.

Tiebreaker 3.2

Tiebreaker 3.2

In the grid below, how many paths from point A to point B are there, if you must stay on the gridlines and are only allowed to move down and to the right?



Rank 1

Score: 11

Rank 1

Score: 11

Winner: **Aryan Raj**

Feedback Form

- Please take a few minutes to fill this out!

Thank You!

- Special thanks to our volunteers and our chaperones for making this happen, and to everyone for showing up!